

JSON-Based Sequencer and Customizable Assets Specification Proposal

Overview

This specification defines how the `sequence.json` drives the immersive experience, managing localized audio, visuals, and text instructions, while keeping static UI assets (like the panel background and welcome screen graphics) in a separate, clearly structured folder for allowing the customization of the visual appearance of the app.

The **Sequencer**, as defined by the `sequence.json` file, must be capable of the following:

- Sending **Arduino commands**
- Displaying and hiding **images or videos** (one at a time only, with preferred resolution)
- Showing/hiding **UI panels and instructional text**
- Playing localized **audio** in **4 languages** (de, en, fr, it)

On the other hand, the following **Static assets** can be defined :

- Panel **background** image
- **Start button** image (on/off states)
- **Language buttons** images
- The **font** used in UI panels and its **color**
- A `config.json` for the languages that will actually be picked

Sequence Step Format

Each timeline event is represented as a step in the JSON. The format for each step is as follows:

```
{
```

```
"time": 0.0,  
"textKey": "Welcome",  
"audio": "welcome.mp3",  
"visual": "instructions.png"  
"actions": []  
}
```

Field Descriptions:

- **time**: Time (in seconds) since the experience starts (triggered when both users are ready).
- **textKey**: A key that maps to a localized instruction string. Also shows the text panel if not already showing.
- **audio**: File name of the audio (.ogg) instruction. Separate files must exist for each language in their respective folders
- **visual**: Image (.png) or video (.mp4) — Loads the visual, use [ShowVisual](#) to show it
- **actions**: List of actions to perform at this step.

Supported Actions

You can include one or more of the following actions in any step:

- [ShowVisual](#)
- [HideVisual](#)
- [HidePanel](#)
- [WallOn](#)
- [WallOff](#)
- [MirrorOn](#)
- [MirrorOff](#)
- [StartExperience](#)
- [EndExperience](#)

Language Support

- Language selection uses country codes: [en](#), [de](#), [fr](#), [it](#), etc.
- Audio is stored in [/Sequence/Audio/{lang}/](#)

- Translation JSONs in `/Sequence/Translations/{lang}.json`
- Flags in `/Static/` using `flag_{code}.png`
- The available languages are auto-detected based on the folder structure
- We must have at least an audio folder, a translation json, and a flag icon for a language to be shown in the interface. The maximum is 4 languages.
- The languages shown are listed in `config.json`
`"visible_languages": ["en", "fr", "it"]`

Folder Structure

Make sure the following folder structure is followed to ensure all the assets are loaded correctly :

```

/TheMachineToBeAnother/
├─ TheMachineToBeAnother.exe
├─ /Content/
│   ├─ config.json
│   ├─ /Sequence/
│   │   ├─ sequence.json
│   │   ├─ Audio/
│   │   │   ├─ en/
│   │   │   │   └─ welcome.ogg
│   │   │   ├─ de/
│   │   │   │   └─ welcome.ogg
│   │   │   ├─ fr/
│   │   │   │   └─ welcome.ogg
│   │   │   └─ it/
│   │   │       └─ welcome.ogg
│   │   └─ Images/
│   │       └─ instructions.png
│   │   └─ Videos/
│   │       └─ instructions.mp4
│   └─ Translations/
│       ├─ en.json
│       ├─ de.json
│       ├─ fr.json
│       └─ it.json
└─ /Static/

```

- | └─ panel_background.png
- | └─ start_button_on.png
- | └─ start_button_off.png
- | └─ flag_en.png
- | └─ flag_de.png
- | └─ flag_fr.png
- | └─ flag_it.png
- | └─ font_color.txt